

Characterization of Plant Accumulation of Pharmaceuticals from Soils with Their Concentration in Soil Pore Water

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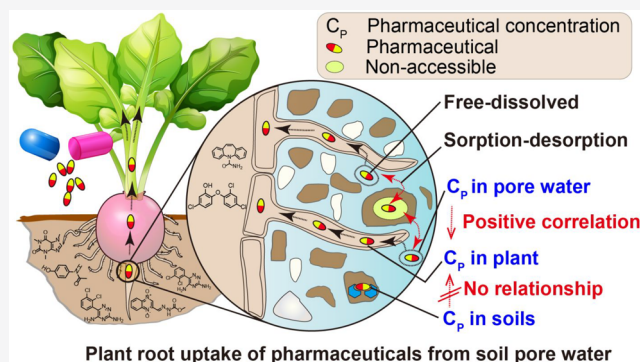
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ABSTRACT: Predicting plant uptake of pharmaceuticals from soils is very challenging because many pharmaceuticals are ionizable compounds, which experience highly variable sorption/desorption and transformation processes in soils. This study aimed to elucidate how the equilibrium between sorbed and dissolved phases influences radish uptake of 15 pharmaceuticals from three soils with different properties. After 30 days of uptake, the accumulation of acetaminophen, carbamazepine, lamotrigine, carbadox, trimethoprim, and triclosan in radish ranked as Riddles > Capac > Spinks soil. In contrast, radish accumulation of caffeine, lincomycin, monensin, tylosin, sulfadiazine, and sulfamethoxazole exhibited the opposite order of Riddles < Capac < Spinks soil. Oxytetracycline and estrone demonstrated similar accumulation in radish grown in the three soils. Accumulation of pharmaceuticals in radish demonstrated no apparent relation with their concentration in soils. However, we identified strong positive correlation between pharmaceutical accumulation in radish and their corresponding concentration in soil pore water. These results reveal that pharmaceutical in soil pore water is the dominant fraction bioavailable to plant uptake. Relatively constant root concentration factors (RCFs) on the basis of pharmaceutical concentration in soil pore water, compared to the highly variable RCFs derived from soils, suggest that pore water-based RCF is superior for describing pharmaceutical accumulation in plants grown in soils. We recommend that pharmaceuticals in soil pore water should be evaluated and included in modeling their uptake by plants.

KEYWORDS: plant uptake, pharmaceuticals, soil pore water, sorption, bioaccumulation



Plant root uptake of pharmaceuticals from soil pore water

INTRODUCTION

Pharmaceuticals have been widely disseminated in agricultural systems via irrigation with treated wastewater and land application of biosolids and animal manures.^{1–3} The resultant pharmaceutical levels in soils range from $\mu\text{g}/\text{kg}$ to low mg/kg .^{4–6} The presence of pharmaceuticals in agricultural lands has led to increased concern about the potential to accumulate in the edible parts of food crops under typical agronomic practices.^{7–13} In most cases, risks associated with human exposure to pharmaceuticals through food consumption are estimated to be low on the basis of acceptable daily intake.^{9,12–15} However, the effects of chronic exposure to a mixture of bioactive pharmaceuticals at low levels is still unclear.^{15,16} A better understanding of plant uptake of pharmaceuticals from soil and water is critical to ensure food safety when utilizing pharmaceutical-contaminated byproducts, such as wastewater, biosolids, and manures in agricultural production.

Soil pore (interstitial) water is believed to act as solvent to desorb pharmaceuticals from soil and as carrier to transport them to crops. The freely dissolved pharmaceuticals in soil pore water are assumed to be bioavailable to plant root uptake.

The relationship between pharmaceutical concentration in soil pore water and accumulation in plant was observed in several previous studies. For example, Carter et al.¹⁷ reported a positive relationship between carbamazepine and verapamil concentration in soil pore water and their accumulation in zucchini leaves. In fact, pharmaceuticals present in soils include both soil-sorbed and pore water-dissolved fractions; the latter represents the primary bioavailable fraction to plant root uptake. Unlike many lipophilic organic contaminants, most pharmaceuticals contain multiple polar and/or ionic functional groups, thus are relatively hydrophilic (e.g., $\log K_{\text{OW}} < 3$) and readily dissolved in soil pore water.^{18,19} Chemicals in soil pore water have been considered as the bioavailable fraction for plant uptake from soils for years,^{20–22} however, there are still large data gaps for quantitative elucidation of the underlying

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Table 1. Physicochemical Characteristics of the Three Studied Soils

soil series	soil texture	pH	organic carbon (%)	CEC (cmol/kg)	clay (%)	silt (%)	sand (%)
Riddles	sandy loam	8.6	1.3	7.0	8.9	17.2	73.9
Capac	sandy clay loam	8.8	2.8	15.2	23.9	21.4	54.7
Spinks	loam	8.3	4.9	23.1	24.8	40.3	34.9

assumption because chemical concentration in soil pore water was rarely measured in the majority of plant uptake studies. Many previous studies only measured pharmaceutical concentration in bulk soils, and reported soil-based root concentration factors (RCF_{soil}).^{9,23–27} Measurement of pharmaceutical concentration in soils, however, cannot well account for the actual amount of pharmaceuticals available to plant uptake.²⁸ The RCF_{soil} values reported from the previous studies with different soils are highly variable,^{29–31} suggesting that RCF_{soil} values measured in a specific set of soils are not sufficient to accurately predict plant uptake of pharmaceuticals from other soils. This is because desorption or the bioavailable fraction of pharmaceuticals is apparently different in the soils of varying characteristics. In a recent study, we identified that pore water-based root concentration factor ($RCF_{\text{pore water}}$), rather than RCF_{soil} , is more appropriate for evaluating the relationship between pharmaceutical physicochemical properties and their accumulation in plant.³⁰ However, only one soil was involved in the experiment. Whether $RCF_{\text{pore water}}$ can be applied to characterize the bioaccumulation potential of pharmaceuticals in plants from different soils has not been well studied. Therefore, more systematic studies are critically needed to evaluate the role of pore water in plant uptake of pharmaceuticals in various soil–plant systems.

Sorption process controls the distribution of pharmaceuticals between soil and pore water. For a given pharmaceutical, the soil–water distribution coefficient (K_d) can vary by 1–2 orders of magnitude in different types of soils.^{32,33} For example, K_d values of carbamazepine, trimethoprim, atenolol, clarithromycin, and sulfamethoxazole in 13 soils can vary up to 18, 24, 24, 26, and 69 times, respectively.³² Soils with different mineralogy, organic matter, cation exchange capacity, or pH can result in a wide range of affinity to pharmaceuticals.^{32,33} Biotic and abiotic transformation of pharmaceuticals also influence the persistence of pharmaceuticals in soils.³⁴ Sorption–desorption and transformation of pharmaceuticals in soil systems are dynamic processes, which combined to regulate the available amount of pharmaceuticals in soil pore water to plant uptake. For instance, the accumulation of carbamazepine, lamotrigine, sulphapyridine, and metoprolol was higher in vegetables grown in a sandy soil than the plants in clay loam and sandy loam soils.³⁵ This could be attributed to weak sorption of pharmaceuticals by the sandy soil, which correspondingly releases a higher pharmaceutical concentration in pore water. Although some experimental results suggest that pharmaceuticals in soil pore water could be the predominant fraction for plant uptake,^{30,31} the direct evaluation on plant uptake via integrating sorption and transformation processes in soil pore water–plant systems is still limited.

This study aimed to test the hypothesis that pharmaceutical uptake by plants is strongly associated with their presence in soil pore water. We evaluated the role of soil pore water in radish (*Raphanus sativus*) uptake and accumulation of 15 pharmaceuticals with varying physicochemical properties (Table S1) from three soils. Greenhouse experiments were

conducted to examine the distribution of the 15 pharmaceuticals in different soil pore water–plant systems. The measurements of pharmaceutical concentration in radish shoots, roots, soils, and soil pore water, and the comparison of RCF_{soil} and $RCF_{\text{pore water}}$ were systematically analyzed. The results of this study provide the key information in developing predictive models on the fate and plant uptake of pharmaceuticals from soils, and assessing the potential risks associated with human exposure via consumption of food crops tainted with pharmaceutical residues.

MATERIALS AND METHODS

Chemicals. The studied pharmaceuticals were selected based on their frequency and concentration detected in treated wastewater, biosolids, animal manure, and agricultural lands,³⁶ and covered a wide range of physicochemical properties and purpose of use (Table S1). Fifteen pharmaceuticals were selected including nine antibiotics (sulfadiazine, sulfamethoxazole, carbadox, triclosan, trimethoprim, lincomycin, oxytetracycline, monensin, and tylosin), two antiepileptics (carbamazepine and lamotrigine), two hormones (estrone and 17 β -estradiol), one antipyretic (acetaminophen), and one stimulant (caffeine). These chemicals were purchased from Sigma-Aldrich (St. Louis, MO, USA). Ceramic homogenizers and dSPE sorbents (octadecylsilane (C18) and primary secondary amine (PSA)) were purchased from Agilent Technologies (Santa Clara, CA, USA). HPLC-grade solvents (methanol and acetonitrile) and reagent-grade acetone were purchased from Fisher Scientific (Fair Lawn, NJ). Ultrapure water was obtained from a laboratory Milli-Q water purification system (Millipore, Billerica, MA, USA).

Soils. To examine the influence of soil properties on the distribution and plant uptake of pharmaceuticals, three soils (Riddles sandy loam, Capac sandy clay loam, and Spinks loam) with a range of organic carbon content and different soil texture were collected from the top 15 cm depth at three sites of Michigan State University Research and Teaching Farm, located in Lansing, Michigan, US. The soils were air-dried and then passed through a 2 mm sieve prior to use. The soil texture and properties were analyzed by Soil and Plant Nutrient Laboratory, Michigan State University (East Lansing, MI) using standard procedures. The selected soil properties are summarized in Table 1.

Greenhouse Experiment. Plant uptake experiments were conducted in a climate-controlled greenhouse using natural sunlight during the summer. Temperature was 25 ± 5 °C during daytime and at 21 ± 5 °C during night. One kilogram of each soil was spiked with a mixture of the 15 pharmaceuticals in acetone and mixed thoroughly using a spatula in a fume hood to reach the initial concentration of 5000 ng/g for each compound. After acetone was evaporated, the spiked soil was mixed thoroughly with 9.0 kg of soil in a clean motorized concrete mixer, resulting in a nominal concentration of about 500 ng/g for each pharmaceutical. Five radish (*Raphanus sativus*) seeds (Burpee & Co., Warminster, PA) were directly sowed in each plastic pot

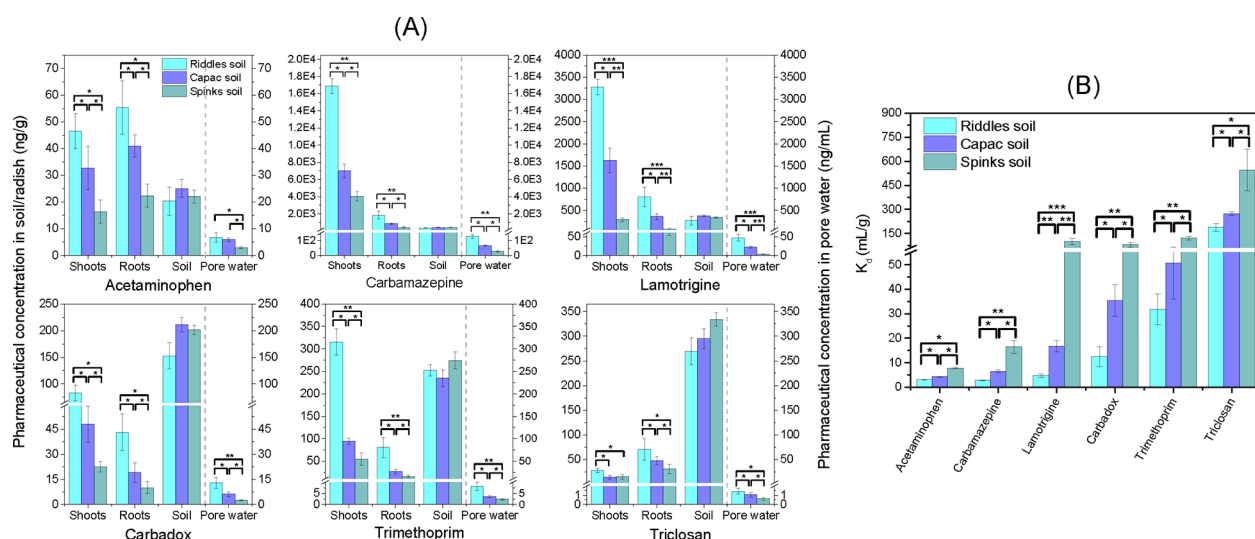


Figure 1. (A) Acetaminophen, carbamazepine, lamotrigine, carbadox, trimethoprim, and triclosan concentration in radish tissues (roots and shoots, ng/g dry weight, left scale), soil (ng/g, dry weight, left scale), and pore water (ng/mL, right scale); (B) soil–pore water distribution coefficient (K_d , mL/g) of acetaminophen, carbamazepine, lamotrigine, carbadox, trimethoprim, and triclosan in the radish-free soil controls. All samples were collected at day 30. Error bars are standard deviations ($n = 3$). *, **, and *** represent the significant difference at $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

(14.6 cm in diameter and 10.8 cm in height) containing ~ 1.0 kg of soil. The pots were prepared in triplicate for each treatment including pharmaceutical-free and radish-free controls. The soil water content was maintained at 60% of soil maximum water holding capacity on a daily basis by adding the same amount of water lost from evapotranspiration. This practice not only provided sufficient amount of water for plant growth, but also warranted no water leaching out from the bottom of the pots. After seed germination, the plants were thinned to two seedlings per pot.

Radish plants were sampled at the mature stage (30 days after sowing). The radish plants were thoroughly rinsed with deionized water to remove attached soil particles, blotted dry with tissue paper, and separated into roots (including the edible bulb fraction) and shoots. Plant samples were weighed, cut into small pieces, freeze-dried and ground to fine powders. At the same time, bulk soil samples were also collected from the planted pots to quantify pharmaceuticals in soils and pore water. Soil pore water (i.e., soil liquid phase) was extracted immediately using the protocol described by Carter et al.³⁷ Briefly, duplicate soil samples (25 g) were placed in two 20 mL disposable plastic syringes, with glass wool plug inserted to the outlet of the syringes. The syringes were placed into 50 mL centrifuge tubes and centrifuged at 7650 g for 20 min. The resulting pore water collected from both centrifuge tubes (~ 3 mL) was passed through a 0.22 μm polytetrafluoroethylene (PTFE) membrane prior to analysis. An aliquot of the pore water (500 μL) was diluted to 1000 μL with methanol in a glass vial prior to analysis.

To quantify the attenuation of pharmaceuticals in soils, soil samples were collected from the radish-free controls 4 h after the daily irrigation for a period of 100 days. The samples were taken at day 0, 4, 9, 15, 20, 25, 30, 35, 70, and 100. The pore water samples were collected and analyzed as described above. The extraction of soil samples was described in the following subsection.

Sample Extraction and Analysis. Radish and soil samples were extracted using a modified QueChERS method

developed by Chuang et al.³⁸ Dry soil samples (2.5 g) or ground radish tissues (0.5 g) were placed in 50 mL polypropylene centrifuge tubes, mixed with 2.0 mL of 150 mg L^{-1} of Na_2EDTA , along with two pieces of ceramic homogenizer. The samples were vortexed for 1 min and extracted with 5.0 mL of acetonitrile/methanol mixture ($v/v = 65/35$) by vigorously shaking for 3 min. Then, Na_2SO_4 (2.0 g) and NaCl (1.0 g) were added, and vortexed for another 2 min. The tubes were centrifuged at 2990 g for 10 min, and 1.3 mL of the supernatant was transferred into 1.5 mL centrifuge tubes containing 250 mg of Na_2SO_4 and d-SPE sorbents (25 mg of C18 and 25 mg of PSA). The samples were vortexed for 1 min, and centrifuged at 9240 g for 10 min. The supernatant was filtered through a 0.22- μm PTFE membrane and stored at -20°C prior to analysis.

The samples were analyzed using a Shimadzu prominence high-performance liquid chromatograph (Columbia, MD, USA) coupled to Sciex 4500 triple quadrupole mass spectrometer (Foster City, CA, USA). The chromatographic separation of pharmaceuticals was achieved by an Agilent Eclipse Plus C18 column (50 mm $\times$ 2.1 mm, particle size 5 μm) (Santa Clara, CA, USA). The detailed LC–MS/MS analysis and the quality assurance and control can be found in Li et al.³⁹ To account for the potential loss of pharmaceuticals, method blank (methanol), matrix blank, matrix spike, and matrix spike duplicate were included in all batches of samples (every 20 samples as a batch). The external matrix-matched standard curves were utilized for quantification to correct the potential matrix effects from soil and plant extracts. The extraction recoveries of spiked pharmaceuticals in soil and plant samples ranged from 56.4% to 111.4% (Table S2), and the method detection limits were within the range of 0.5–6.5 ng/g dw (or ng/mL) (summarized in Table S3).

Statistical Analysis. One-way analysis of ANOVA ($\alpha = 0.05$) was employed to assess the significant difference for the data in radish or in soils. Statistical analysis was conducted using SPSS 22.0 software for Windows (IBM Corp., Armonk, NY).

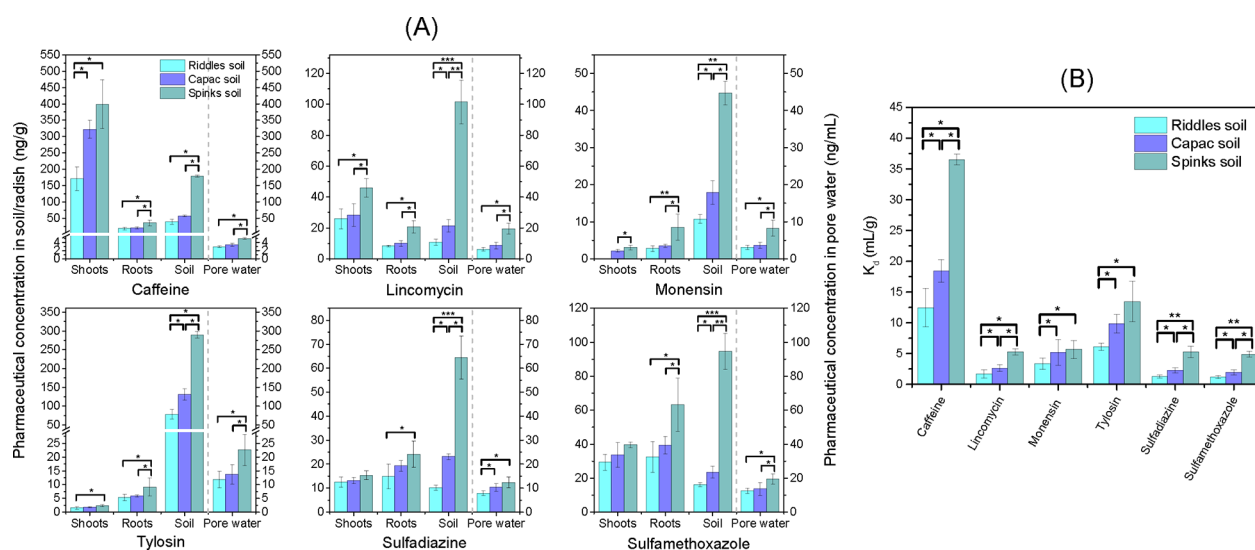


Figure 2. (A) Caffeine, lincomycin, monensin, tylosin, sulfadiazine, and sulfamethoxazole concentration in radish tissues (roots and shoots, ng/g dry weight, left scale), soil (ng/g, dry weight, left scale), and pore water (ng/mL, right scale); (B) soil–pore water distribution coefficient (K_d , mL/g) of caffeine, lincomycin, monensin, tylosin, sulfadiazine, and sulfamethoxazole in the radish-free soil controls. All samples were collected at day 30. Error bars are standard deviations ($n = 3$). *, **, and *** represent the significant difference at $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

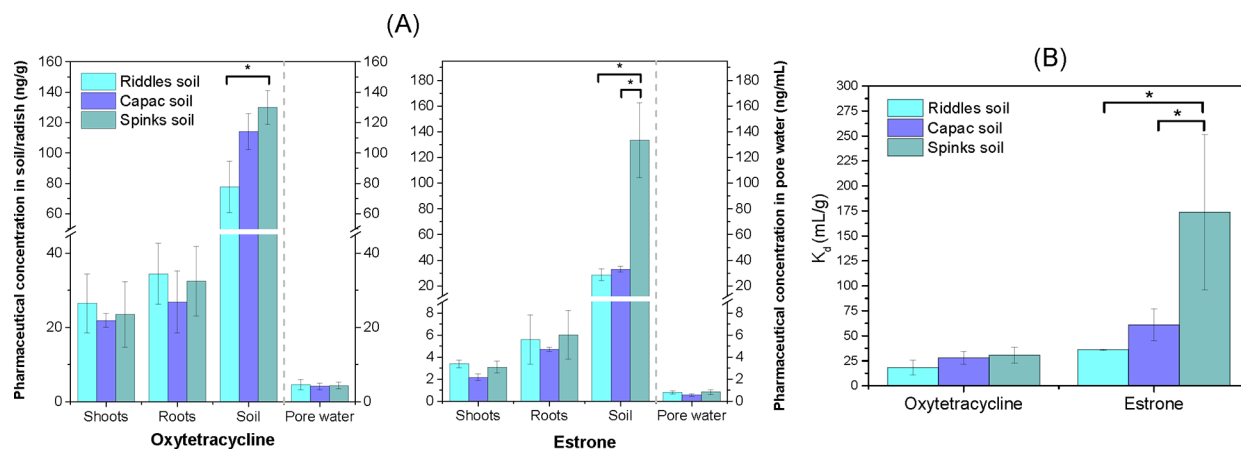


Figure 3. (A) Oxytetracycline and estrone concentration in radish tissues (roots and shoots, ng/g dry weight, left scale), soil (ng/g, dry weight, left scale), and pore water (ng/mL, right scale); (B) soil–pore water distribution coefficient (K_d , mL/g) of oxytetracycline and estrone in the radish-free soil controls. All samples were collected at day 30. Error bars are standard deviations ($n = 3$). * represent the significant difference at $p < 0.05$.

RESULTS AND DISCUSSION

Soil Pore Water Carries Pharmaceuticals to Radish.

The spiked pharmaceutical concentration in soils before the start of greenhouse experiments ranged from 283 to 504 ng/g (Table S4), and there was no significant difference for a given pharmaceutical in the three soils ($p > 0.05$). The pharmaceutical concentration in radish tissues (roots and shoots), pore water, and three soils after 30 days of cultivation is shown in Figures 1–3. Fourteen out of the 15 pharmaceuticals (except 17 β -estradiol) were found to accumulate in radish. No accumulation of 17 β -estradiol was likely due to its extensive metabolism in radish.³⁹ Pharmaceuticals in soils underwent varying degrees of sorption and/or transformation (Table S5), and their accumulation in radish grown in different soils also varied. To better elucidate the uptake results, we divided the 14 pharmaceuticals into three categories: Group 1, no significant difference in pharmaceutical concentration in soils ($p > 0.05$) but different accumulation in radish ($p < 0.05$) (Figure 1); Group 2, increase in

pharmaceutical accumulation in radish with increasing concentration in soils (Figure 2); Group 3, no difference of the accumulation in radish ($p > 0.05$), but different concentration in soils ($p < 0.05$) (Figure 3).

The first group of pharmaceuticals included acetaminophen, carbamazepine, lamotrigine, carbadox, trimethoprim, and triclosan. After 30 days of greenhouse experiment, the concentration of these six pharmaceuticals in the three studied soils (C_{soil}) had no significant difference ($p > 0.05$) (Figure 1A), suggesting that their biotic and abiotic dissipation was not significantly impacted by soil types during the experimental period. However, pharmaceutical concentration in radish roots (C_{root}) and shoots (C_{shoot}) decreased in the order of Riddles soil > Capac soil > Spinks soil, except that no significant difference ($p > 0.05$) was observed for triclosan in radish shoots between Capac and Spinks soils. The accumulation of lamotrigine in radish tissues grown in Riddles soil was >10 times more than that in Spinks soil. It is noted that the concentration of these pharmaceuticals in soil pore water (C_{pw}) decreased in the order of Riddles > Capac > Spinks soil,

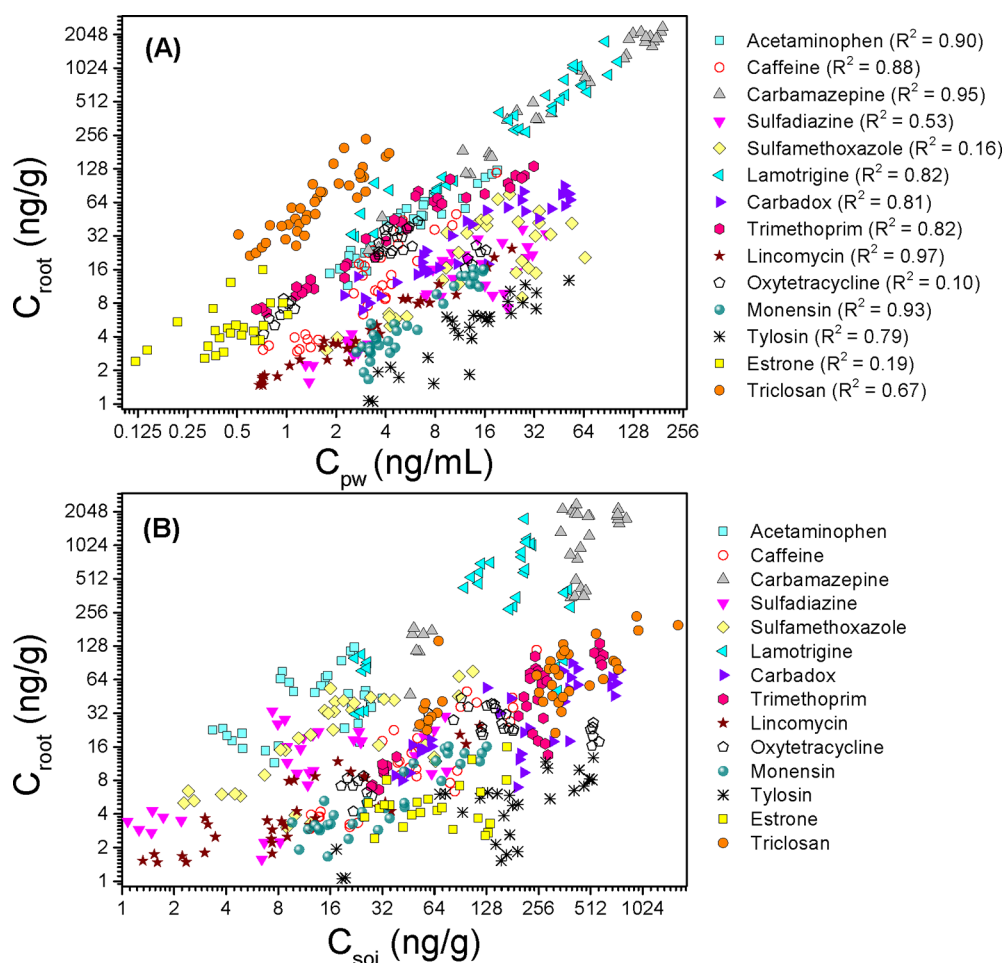


Figure 4. (A) Relation between pharmaceutical concentration in radish roots (C_{root}) and their concentration in soil pore water (C_{pw}). (B) Relation between C_{root} and pharmaceutical concentration in bulk soil (C_{soil}). For each pharmaceutical there are the total of 33 data pairs including $n = 9$ from this study, plus the remaining from Li et al. (2019 and 2020), and Fu et al. (2016b), except for tylosin $n = 27$, estrone $n = 24$, and triclosan $n = 37$. n is the number of data pairs present in this figure.

which is in the same order as their accumulation in radish roots and shoots (Figure 1A). The greater accumulation of pharmaceuticals in radish grown in Riddles soil than that in Capac soil or Spinks soil could be attributed to the higher sorption by the latter two soils, reducing pharmaceutical concentration in the pore water. In this study, we used the measurements of the radish-free controls to calculate the distribution coefficient of pharmaceuticals (K_d) between soil and pore water ($K_d = C_{\text{soil}}/C_{\text{pw}}$). The K_d values of the six pharmaceuticals increased in the order Riddles < Capac < Spinks soil (Figure 1B). The organic carbon content and clay fraction were 1.3% and 8.9% in Riddles soil, 2.8% and 23.9% in Capac soil, and 4.9% and 24.8% in Spinks soil, respectively (Table 1). The higher soil organic matter content and clay mineral fraction could lead to greater sorption for pharmaceuticals, and hence reduce their bioavailability.¹⁹ Several previous studies also found that carbamazepine accumulation in cucumber,^{35,40,41} cabbage,⁴² and tomato³⁵ was inversely related to soil organic matter content. In Figure 1A, pharmaceutical concentration in the three soils was not significantly different ($p > 0.05$); however, significant difference in pharmaceutical accumulation in radish was observed ($p < 0.05$). This suggests that pharmaceuticals, even at similar concentration in soils, demonstrated varying bioavailability to plant uptake. Strong sorption of pharmaceuticals by soils could

reduce their concentration in pore water, leading to the decreased bioavailability to plant uptake.

The second group of pharmaceuticals included caffeine, lincomycin, monensin, tylosin, sulfadiazine, and sulfamethoxazole. Their concentration in the three soils showed a significant difference ($p < 0.05$) in the order of Riddles < Capac < Spinks soil after the 30-day experiment (Figure 2A). The K_d values were consistent with the order of pharmaceutical concentration in the soils (Figure 2B). This occurs because the strong sorption of pharmaceuticals by soils might reduce their transformation in soils.¹⁹ The different transformation rates could lead to the varying pharmaceutical concentrations in soils (Table S5). In contrast to the six pharmaceuticals shown in Figure 1, the accumulation of these six pharmaceuticals in radish generally increased with increasing K_d values in the order of Riddles < Capac < Spinks soil (Figure 2). It is noted again that the accumulation in radish followed the same order as the pharmaceutical concentration in soil pore water (Figure 2A). Therefore, pharmaceutical concentration in soil pore water, rather than the concentration in soil, determines the bioavailable fraction of pharmaceuticals to plant uptake.

To further clarify this point, we estimated chemical concentration in soil water as the ratio of C_{soil}/K_d .²¹ In this study, C_{soil} was measured from the radish-grown pot soils, and

K_d value was obtained from the radish-free controls. For example, lincomycin concentration was 10.7 ng/g in the Riddles soil, 22.0 ng/g in the Capac soil, and 101.5 ng/g in the Spinks soil after 30-day exposure (Figure 2A), and the corresponding K_d was 1.7, 2.6, and 5.3 mL/g (Figure 2B). The estimated lincomycin concentration in soil water is 6.2 ng/mL in the Riddles soil, 8.5 ng/mL in the Capac soil, and 19.3 ng/mL in the Spinks soil. These values are comparable to the measured lincomycin concentration in soil pore water from the three soils. A similar trend is also observed for caffeine, monensin, tylosin, sulfadiazine, and sulfamethoxazole. It is believed that the higher pharmaceutical concentration in soil pore water is responsible for the greater accumulation in radish.

The third group of pharmaceuticals included oxytetracycline and estrone. For these two pharmaceuticals, there was no significant difference ($p > 0.05$) in the concentration either in soil pore water or in radish tissues, despite the significant difference ($p < 0.05$) in their concentration in soils (Figure 3A). The similar levels of pharmaceuticals in soil pore water could be responsible for their comparable accumulation in radish. Besides, the K_d values ($C_{\text{soil}}/C_{\text{pw}}$) of the studied pharmaceuticals were comparable in radish-grown pot soils vs in radish-free pot soils (Table S6). The similar K_d values in these two systems suggest that pharmaceutical sorption results in the unplanted soils could be applied to estimate their bioavailability to plant uptake. Together with the above analysis, all results indicate that although sorption/desorption and biotic/abiotic transformation could lead to the varying pharmaceutical concentration in soils, the amount of pharmaceuticals present in soil pore water is most closely associated with the accumulation in radish.

Relation between C_{root} (or C_{shoot}) and C_{pw} vs C_{root} (or C_{shoot}) and C_{soil} . To further examine the role of pore water in pharmaceutical uptake by radish, C_{root} (or C_{shoot}) was plotted against C_{pw} and C_{soil} using the data from the present study, as well as from the previously published studies (Figure 4 and Figure S1).^{29–31} These data included radish uptake of pharmaceuticals at different soil-sorbed concentrations and from biosolids- and biochar-amended soils, which achieved a wide range of pharmaceutical concentrations in the pore water.^{29,31} As shown in Figure 4A, a strong positive relationship was observed between C_{root} and C_{pw} for acetaminophen ($R^2 = 0.90$), caffeine ($R^2 = 0.88$), carbadox ($R^2 = 0.81$), carbamazepine ($R^2 = 0.95$), lamotrigine ($R^2 = 0.82$), lincomycin ($R^2 = 0.97$), monensin ($R^2 = 0.93$), sulfadiazine ($R^2 = 0.53$), trimethoprim ($R^2 = 0.82$), triclosan ($R^2 = 0.67$), and tylosin ($R^2 = 0.79$) (Table S7). Similarly, the positive correlation was also found between C_{shoot} and C_{pw} for these pharmaceuticals, with R^2 values between 0.54 and 0.95 (Figure S1A, Table S8). These results clearly indicate that the potential uptake of pharmaceuticals by radish plants is directly related to pharmaceuticals present in soil pore water originating from desorption from soils. In addition, the linear relationship suggests that pharmaceutical uptake by radish from soils via pore water could be a passive process driven by the concentration gradient between soil pore water and plant roots. The relatively weak linear relationship between C_{root} (or C_{shoot}) and C_{pw} for oxytetracycline, estrone, and sulfamethoxazole might be due to their extensive metabolism in radish.³⁹

In contrast, the relationship between C_{root} (or C_{shoot}) and C_{soil} for the studied pharmaceuticals demonstrated more random distribution (Figures 4B and S1B; Tables S7 and

S8). Goldstein et al.³⁵ also showed no specific relationship between pharmaceutical concentration in plants (cucumber and tomato) and in soils for caffeine, carbamazepine, and lamotrigine. The strong positive correlations between C_{root} (or C_{shoot}) and C_{pw} vs no apparent relations between C_{root} (or C_{shoot}) and C_{soil} indicate that soil heterogeneity (e.g., microbial communities, organic matter, mineral content) could influence the sorption and transformation processes of pharmaceuticals in soils, which subsequently affect their actual bioavailable concentration (C_{pw}) to root uptake. These results were observed without surprise, considering that plants take up pharmaceuticals primarily from soil pore water. Soil-sorbed pharmaceuticals could be released to pore water at different extent depending on their affinity to soils. The abiotic and biotic transformations in soil–water systems also influence the amount of pharmaceuticals present in pore water and soil. However, in all cases, the pharmaceutical accumulation in radish roots/shoots has been shown to be positively related to the pharmaceuticals present in soil pore water, but not related apparently to their concentration in soils. Therefore, we strongly argue that pharmaceutical concentration in soil pore water should be measured when evaluating plant uptake of pharmaceuticals from soils. To the best of our knowledge, there are still very limited data of the actual measurements of pharmaceutical concentration in soil pore water. This is probably due to technical difficulties in soil pore water sampling (e.g., special sampling devices, long sampling time, lacking standard procedure, and low soil moisture content).⁴³ The centrifugation method adopted here and in several previous studies provides a simple, inexpensive but effective technique to collect soil pore water, which could be used to measure the bioavailable fraction of pharmaceuticals from soils to plant uptake.^{17,31,37}

RCF_{soil} vs $\text{RCF}_{\text{pore water}}$. To better elucidate the influence of soil sorption on pharmaceutical accumulation in radish, RCF values were calculated on the basis of both C_{soil} ($\text{RCF}_{\text{soil}} = C_{\text{root}}/C_{\text{soil}}$) and C_{pw} ($\text{RCF}_{\text{pore water}} = C_{\text{root}}/C_{\text{pw}}$) (Figure 5). For the studied pharmaceuticals (except monensin), RCF_{soil} values demonstrated significant difference ($p < 0.05$) between any two soils (Figure 5A), and the values decreased in order of Riddles > Capac > Spinks soil, except for oxytetracycline which followed Riddles (0.45 g/g) > Capac (0.24 g/g) $\approx$ Spinks soil (0.25 g/g). Lamotrigine had the greatest difference in RCF_{soil} values ranging from 0.23 g/g in Spinks soil to 3.7 g/g in Riddles soil. For the three soils, the measured soil pH values were within a narrow range between 8.3 and 8.8. Thus, the influence of pH on the variation of RCF_{soil} values among the three soils is likely minimal, because the pK_a of the pharmaceuticals is away from this pH range (Table S1). Therefore, the decreasing RCF_{soil} values from Riddles to Capac to Spinks soil can be attributed to the decreased bioavailability of soil-sorbed pharmaceuticals. The strong sorption by Spinks soil reduced the bioavailable fraction of pharmaceuticals in the soil, that is, the less amount of pharmaceutical was desorbed to pore water, leading to the decreased uptake by radish. The varying bioavailability of pharmaceuticals in soils limits the accurate prediction of uptake potential by plant from different soils due to the wide range of sorption affinities, when using RCF_{soil} measured for a specific soil.

Unlike the RCF_{soil} values, $\text{RCF}_{\text{pore water}}$ values remained more constant, e.g., no significant difference ($p > 0.05$) for a given pharmaceutical among the three soils (Figure 5B). As expected, the variation for leaf concentration factor based on

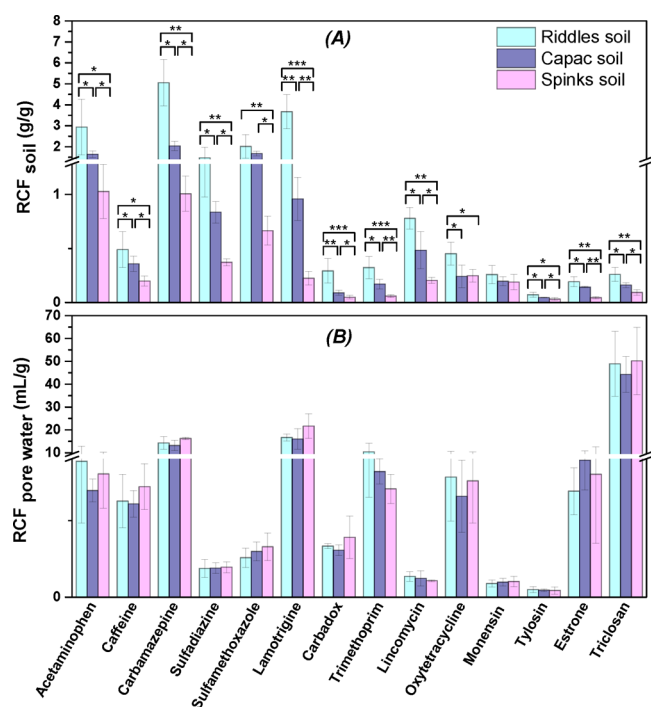


Figure 5. (A) Soil-based root concentration factor (RCF_{soil}) and (B) pore water-based root concentration factor ($RCF_{pore\ water}$) of pharmaceutical accumulation by radish from soils. Error bars are standard deviations ($n = 3$). *, **, and *** represent the significant difference of RCF_{soil} values between two soils at $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

pharmaceutical concentration in pore water ($LCF_{pore\ water}$) was also much less than the LCF values based on the concentration in soils (LCF_{soil}) (Figure S2). These results indicate that $RCF_{pore\ water}$ (or $LCF_{pore\ water}$) is a more reasonable parameter to evaluate plant bioaccumulation potential of pharmaceuticals from soils. More importantly, the consistency of $RCF_{pore\ water}$ (or $LCF_{pore\ water}$) in radish roots (or shoots) further confirms that the freely dissolved pharmaceuticals in soil pore water represent the dominant bioavailable fraction for plant uptake. In this manner, the measurement of $RCF_{pore\ water}$ (or $LCF_{pore\ water}$) is highly recommended for characterizing plant accumulation of pharmaceuticals from soils in plant uptake models.

To further confirm the consistency of $RCF_{pore\ water}$ for pharmaceuticals among various soils, triclosan was selected as an exemplar. We analyzed the uptake data of triclosan from soils by radish reported from the present and previous studies.^{29–31,37,44,45} The soil organic carbon content ranged from 0.4 to 8.0%. As illustrated in Figure 6 (left scale), among the 15 soils, the RCF_{soil} values varied significantly up to 23 times from Spinks soil ($RCF_{soil} = 0.09\text{ g/g}$) to soil A ($RCF_{soil} = 2.1\text{ g/g}$) (all data are provided in Table S9). In contrast, the corresponding $RCF_{pore\ water}$ values remained consistent within a factor of 2.9 among the 11 types of soils (Figure 6 right scale),^{29–31} which has much less variation than the RCF_{soil} values. The data reported by Fu et al.²⁹ did not include triclosan concentration in soil pore water. Instead, the $RCF_{pore\ water}$ was estimated using $RCF_{pore\ water} = RCF_{soil} \times K_d$ in which the K_d values were obtained from Fu et al.⁴⁶ The triclosan $RCF_{pore\ water}$ values ($46.7 \pm 3.1\text{ mL/g}$) obtained from the present study were lower than the values ($70.1 \pm 20.1\text{ mL/g}$) estimated from Fu et al.²⁹ This is due likely to the influence

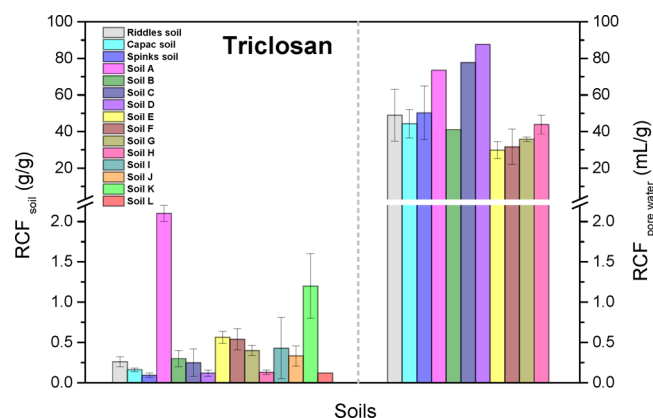


Figure 6. RCF_{soil} (left scale) and $RCF_{pore\ water}$ (right scale) values of triclosan accumulation in radish from 15 soils. The data of Riddles soil, Capac soil, and Spinks soil are obtained from the present study. The data of soil A–L are collected from the previous studies (Carter et al., 2014; Fu et al., 2016b; Li et al., 2019; Li et al., 2020; Pannu et al., 2012; Prosser et al., 2014).

of triclosan speciation at different soil pH values. The soil pH was 6.4–6.6 at which approximately 96% of triclosan was neutral and 4% was anionic species.²⁹ In our study, soil pH values were within 8.3 and 8.8 at which about 17% of triclosan was neutral and 83% was anionic species. The neutral species is favored for root uptake, while the anionic species could be repelled by the negatively charged cell walls and cytosol, and thus inhibit the penetration of anionic molecules through cell membranes.⁴⁷ The lower neutral fraction of triclosan may account for the reduced $RCF_{pore\ water}$ because the neutral species generally has a higher uptake potential. Besides, the substantial in-plant metabolism of triclosan might also influence the variation of accumulation.^{39,48} Nason et al. recently also found that wheat plant favored the uptake of neutral lamotrigine species (a weak base) at higher pH (e.g., 7.4) relative to that at pH 5.7.⁴⁹ For the other 13 pharmaceuticals, the $RCF_{pore\ water}$ values were also within a narrow range of variation, compared to the range of RCF_{soil} values for the radish grown in the soils or soils amended with biochars (Tables S10–S22). The main drawback of RCF_{soil} is the lump of sorption/transformation and root uptake into one parameter, which could substantially limit the utility in plant uptake modeling. In contrast, $RCF_{pore\ water}$ values remain relatively constant and independent of pharmaceutical sorption/transformation processes in soils, and are highly recommended to be used in modeling pharmaceutical uptake by plants from soils.

The pharmaceutical physicochemical properties (e.g., hydrophobicity and ionization) may influence the uptake and bioaccumulation of pharmaceuticals in a soil–plant system. Neutral species of pharmaceuticals generally exhibit a greater uptake by plants than the ionic species, and cationic species usually manifest a higher bioaccumulation than anionic forms.^{7,30,35} The relationship analysis revealed no apparent correlation between $\log RCF_{pore\ water}$ and $\log K_{ow}$ ($R^2 = 0.035$) for all tested pharmaceuticals (encompassing neutral, acid, and basic compounds) in the three soils (Figure S3). However, a strong positive correlation was observed between $\log RCF_{pore\ water}$ and $\log K_{ow}$ ($R^2 = 0.83$) for the pharmaceuticals dominant in neutral species ($n = 7$) (Figure S4). These results suggest that partitioning could be the major driving force for the uptake of neutral pharmaceuticals by plants, while the

accumulation of ionic pharmaceuticals might be governed by other processes such as electrostatic interactions with cell walls and ion trapping.⁴⁷ For example, in the three alkaline soils used in this study (pH = 8.3–8.8), anionic sulfadiazine and monensin (negatively charged species >95%) demonstrated a low bioaccumulation in radish roots ($RCF_{\text{pore water}} < 2.0$ mL/g), because they could be repulsed by the negatively charged root cell membranes.⁴⁷ On the other hand, ion-trapping effects might enhance the bioaccumulation cationic species of pharmaceuticals such as trimethoprim ($RCF_{\text{pore water}} = 7.1$ – 10.4 mL/g) and lamotrigine ($RCF_{\text{pore water}} = 16.1$ – 21.7 mL/g) despite their relatively low hydrophobicity (log K_{ow} of 0.91 and 2.57, respectively). This could be attributed to the speciation alteration and accumulation in cell vacuole with a lower pH $\approx$ 5.5 via ion trap, as well as the electrostatic attraction of the positively charged species to negative charges on cell membranes.⁵⁰

Environmental Implication. It has been hypothesized that chemicals in soil pore water are more appropriate than those in bulk soil to evaluate plant accumulation of contaminants; however, there still remains a lack of substantial experimental results that directly validate this hypothesis, especially for pharmaceuticals. In this study, the linear relationship between the accumulation of pharmaceuticals in radish and their concentration in soil pore water, and the relatively constant $RCF_{\text{pore water}}$ values indicate that pharmaceuticals in soil pore water are the dominant bioavailable fraction for plant root uptake. Sorption/desorption and transformation of pharmaceuticals in soils collectively govern the bioavailable amount of pharmaceuticals in pore water for root uptake. The similar $RCF_{\text{pore water}}$ values, compared to the widely varying RCF_{soil} values of the 14 pharmaceuticals among the three soils, indicate that $RCF_{\text{pore water}}$ could be a very useful parameter for predicting bioaccumulation of pharmaceuticals in plants.

Many pharmaceuticals are ionizable and could manifest multiple charged species in soil pore water.^{51,52} Unlike that of traditional lipophilic organic contaminants, the sorption of polar and ionizable pharmaceuticals to soils is not solely associated with soil organic matter; other parameters such as soil pH, clay minerals, and ionic strength also influence the sorption.¹⁹ Thus, their distribution between soil and pore water cannot simply be estimated by K_{om} . The plant rhizosphere is a complicated ecosystem which impacts the behaviors of pharmaceuticals in soil pore water and hence plant uptake. For example, root exudates could alter the bioavailable fractions of organic contaminants in soil pore water.^{53,54} Plant-induced rhizosphere pH variations could also influence the speciation of ionizable pharmaceuticals in soil pore water and thus the plant uptake.⁴⁵ Therefore, we suggest that pharmaceutical concentration in soil pore water should be measured in order to better evaluate the uptake potential of pharmaceuticals by plants. This is particularly useful to model the fate, transport, and crop uptake of pharmaceuticals in agricultural soils irrigated with treated wastewater and amended with animal manures and biosolids.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.2c00303>.

Physiochemical properties of pharmaceuticals, analytical methods, spiked pharmaceutical concentration in soils, pharmaceutical half-lives in soils, and the detailed RCF_{soil} and $RCF_{\text{pore water}}$ values in radish roots from different soils measured in this study and in the literature (PDF)

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Notes

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